

Companion to Bird & de Moor, Algebra of Programming

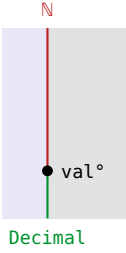
§6. Recursive Programs

§6.1. Digits of a number

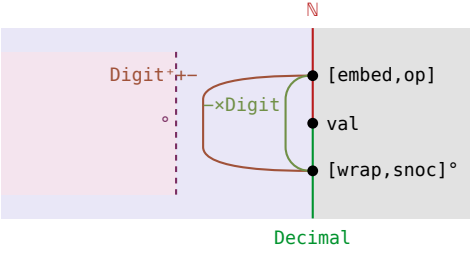
(6.1a)

$val^\circ = embed^\circ wrap \cup op^\circ -xDigit(val^\circ)snoc$

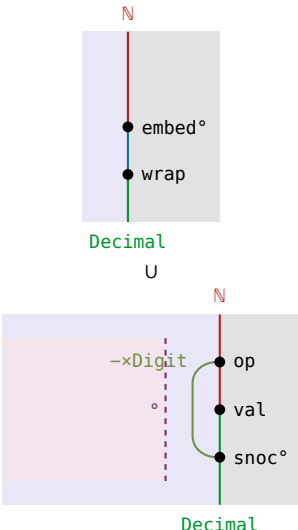
a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest



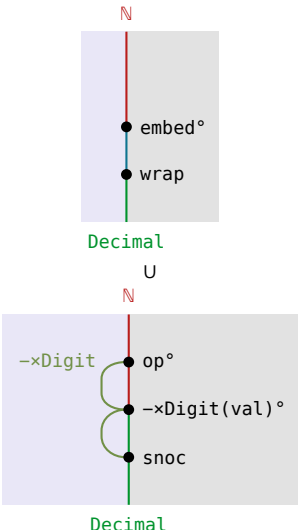
(a)



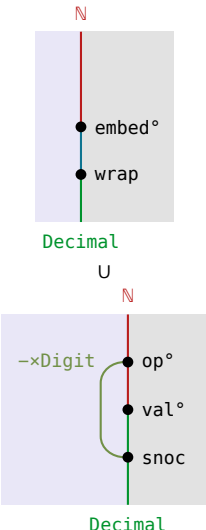
(b)



(c)



(d)



(e)

(a)	val°	the converse of val , which reads digits as a number
(b)	$([wrap, snoc]^\circ Digit^{+-} (-xDigit(val)) [embed, op])^\circ$	val undoes wrap or snoc , reads the front with val and combines by embed or op , so its converse runs that backwards
(c)	$(wrap^\circ embed)^\circ \cup (snoc^\circ -xDigit(val) op)^\circ$	undoing wrap or snoc first gives two cases: a decimal and its last digit read by val then op , or one digit read by embed
(d)	$embed^\circ wrap \cup op^\circ -xDigit(val)^\circ snoc$	the converse of each case is its steps undone in reverse: undo op , undo reading the front, then snoc ; or undo embed , then wrap
(e)	$embed^\circ wrap \cup op^\circ -xDigit(val^\circ) snoc$	undoing the read of the front while keeping the last digit is val on the front alone

$op^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$	(6.1b)
op° splits a number into its quotient and remainder by 10	
$(\exists p. 0 < \pi_1 p \wedge op^\circ(m, p)) \iff 10 \leq m$	
op° gives a pair with a nonzero first component exactly at the numbers with two or more digits	
$(\exists d. embed^\circ(m, d)) \iff 0 < m \wedge m < 10$	
$embed^\circ$ gives a digit exactly at the one-digit numbers	

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

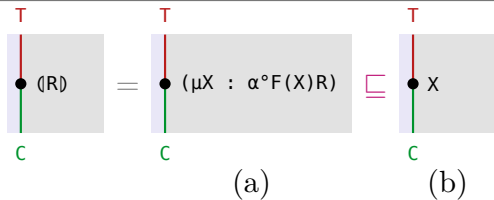
(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

$$\alpha^\circ F(X)R \sqsubseteq X \Rightarrow (R) \sqsubseteq X$$

(6.2c)

a relation X that contains one unfolding of the fold's recursion at X contains the fold



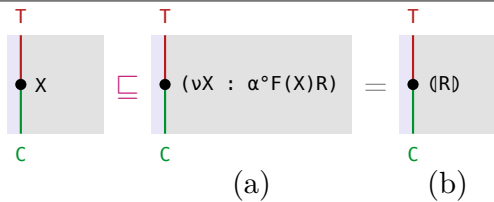
(a) (R) the least fixed point — Theorem 6.1, (6.2b)

(b) $\alpha^\circ F(X)R \sqsubseteq X$, μ below every prefix point

$$X \sqsubseteq \alpha^\circ F(X)R \Rightarrow X \sqsubseteq (R)$$

(6.2d)

a relation X contained in one unfolding of the fold's recursion at X is contained in the fold



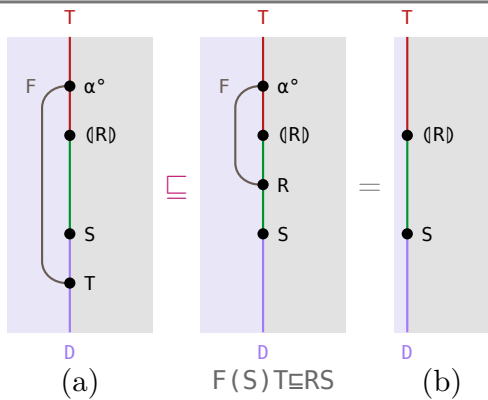
(a) $X \sqsubseteq \alpha^\circ F(X)R$, ν above every postfix point

(b) (R) the greatest fixed point

$$F(S)T \sqsubseteq RS \Rightarrow (T) \sqsubseteq (R)S$$

(6.2e)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S



(a) (6.2) at $X \triangleq (R)S$

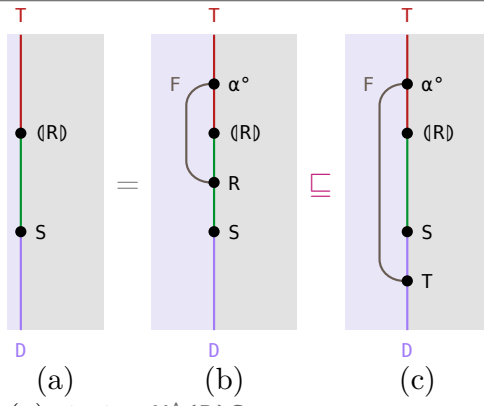
$F(RS) = F(R)F(S)$ — $\triangleq$ relator

(b) $F((R))R = \alpha(R)$, $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce

$$RS \sqsubseteq F(S)T \Rightarrow (R)S \sqsubseteq (T)$$

(6.2f)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



(a) (6.3) at $X \triangleq (R)S$

(b) $\alpha(R) = F((R))R$, $\alpha^0 \alpha = 1$ — $\triangleq$ reduce

(c) $RS \sqsubseteq F(S)T$

$F(R)F(S) = F(RS)$ — $\triangleq$ relator

§6.3. Hylomorphisms

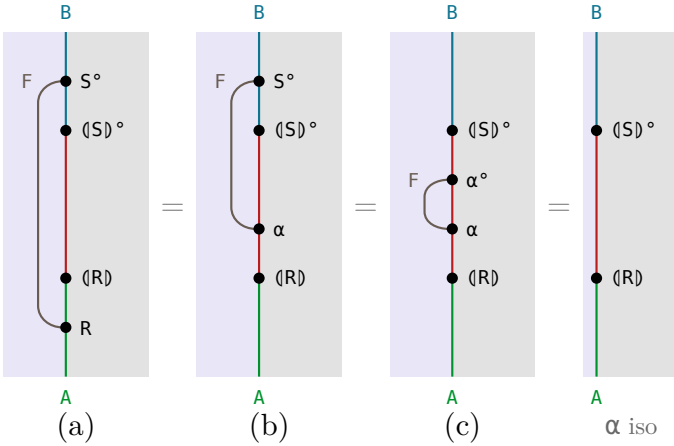
$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : \mathbb{S}^\circ F(X)R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

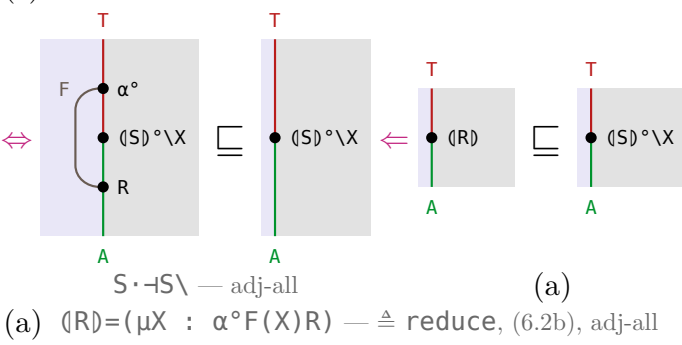
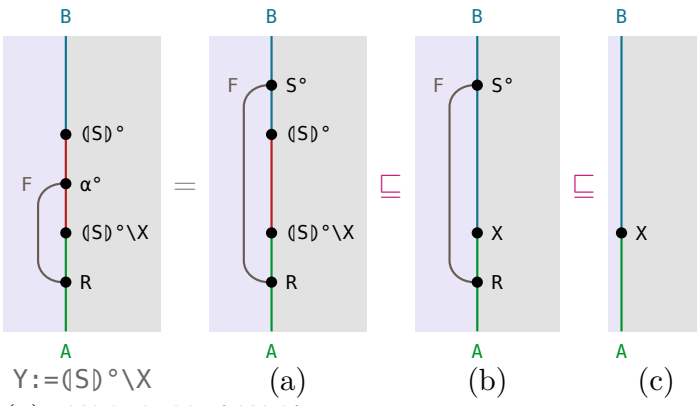
$$\mathbb{S}^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term $\mathbb{S}^\circ$ represents the decomposition stage, $F((\mathbb{S})^\circ(\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $\mathbb{S} : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



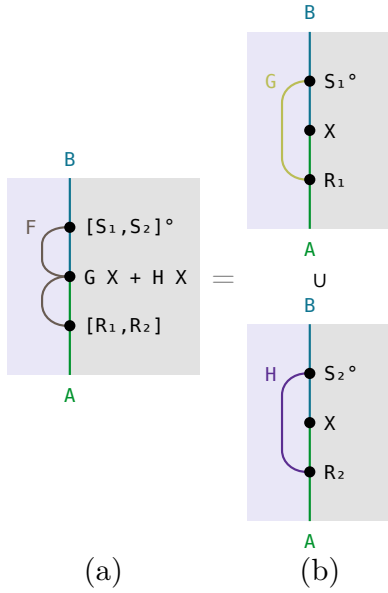
$$\mathbb{S}^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ(\mathbb{R})$ is included in X if X satisfies the associated recursion inequation



$$(\forall x. \forall y. \forall X. F(X)=G(X)+H(X)) \Rightarrow ([S_1, S_2]^\circ \circ ([R_1, R_2])^\circ) = (\mu X : S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

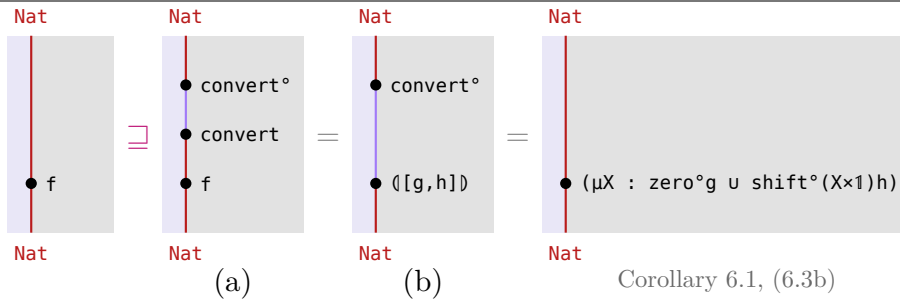


(a)	$[S_1, S_2]^\circ (G(X)+H(X)) [R_1, R_2]$	the body of Theorem 6.2 at $F(X)=G(X)+H(X)$
(b)	$S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2$	$[S, T] = (S+T) \nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$; the $+$ stands for the codiagonals ∇°, ∇ , which are not drawn $\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct

§6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1) h) \sqsubseteq f \quad (6.4a)$$

once **f** sends **zero** to **g** and turns **shift** into **h**, the recursion that halves the argument at each step computes **f**



Corollary 6.1, (6.3b)

- (a) convert simple
(b) fusion: $\text{zero } f = g, \text{shift } f = (f \times 1) h$

$$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes **a** to the power **b**

$\text{zero } \text{exp}(a) = \text{one}$

a to the power 0 is 1

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

a to the power $2n+d$ is $\text{op}(a)$ of **a** to the power **n** and **d**

$$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves **a** at each step computes the remainder of **a** by **b**

$\text{zero } \text{mod}(b) = \text{zero}$

0 mod **b** is 0

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

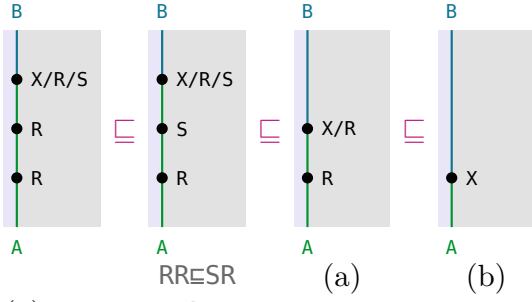
$(2a+d) \text{ mod } b$ is $\text{op}(b)$ of **a** mod **b** and **d**

§6.5. Unique fixed points

$$\text{inductive}(S) \wedge R \sqsubseteq SR \Rightarrow \text{inductive}(R)$$

(6.5a)

R is inductive when $R \sqsubseteq SR$ for an inductive S

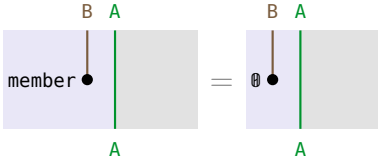


- (a) division by S
 (b) division by R

$$\text{member} = \emptyset$$

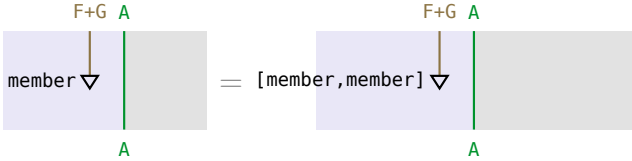
(6.5b)

a constant relator records no elements, so its membership is empty



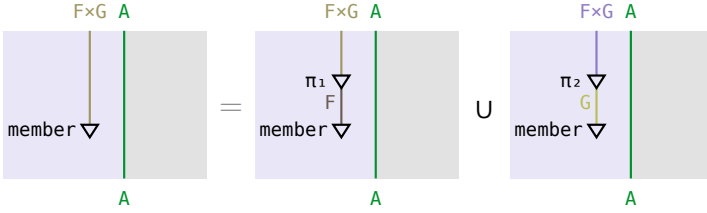
$$\text{member} = [\text{member}, \text{member}]$$

a member of $F+G$ is a member of whichever summand the value is in



$$\text{member} = \pi_1 \text{ member} \cup \pi_2 \text{ member}$$

a member of $F \times G$ is a member of either component



$$\text{inductive}(\text{Smember}) \wedge X = \text{SF}(X)R \wedge Y = \text{SF}(Y)R \Rightarrow X = Y$$

(6.5c)

when S member is inductive, two solutions X, Y of $X = \text{SF}(X)R$ are equal

$$\text{inductive}(\text{Smember}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge \text{SF}(X)R \sqsubseteq X \Rightarrow \text{Entire}(X)$$

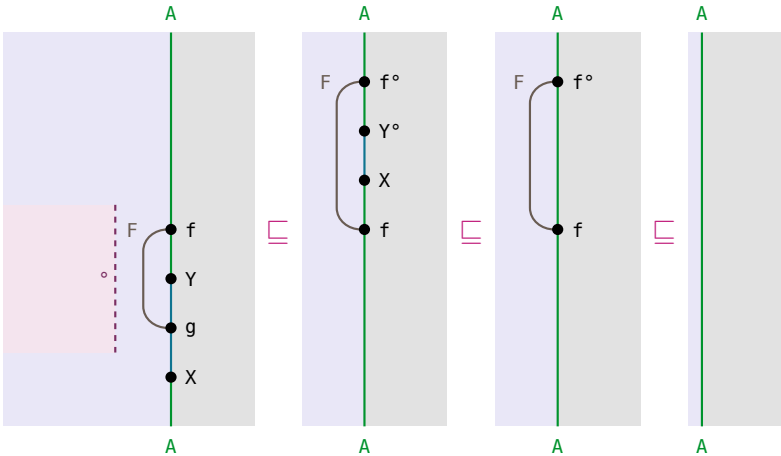
(6.5d)

when S member is inductive and S, R are entire, a pre-fixed point of $X \mapsto \text{SF}(X)R$ is entire

$$\text{inductive}(\text{gmember}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = gF(X)f \Rightarrow \text{Map}(X)$$

(6.5e)

when g member is inductive and f, g are maps, the solution of $X = gF(X)f$ is a map

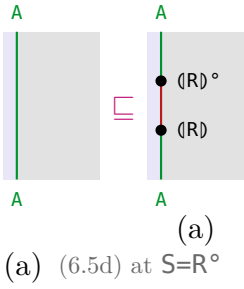


- (a) $X = gF(X)f$, g simple

$$\text{inductive}(\mathbf{R}^\circ \text{member}) \wedge \mathbf{1} \sqsubseteq \mathbf{R}^\circ \mathbf{R} \Rightarrow \mathbf{1} \sqsubseteq (\mathbf{R})^\circ (\mathbf{R})$$

(6.5f)

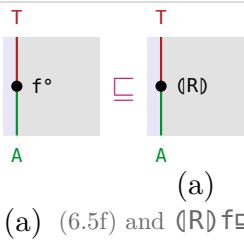
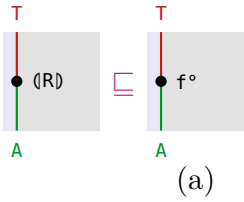
when $\mathbf{R}^\circ \text{member}$ is inductive and $\mathbf{R}$ is surjective, the fold $(\mathbf{R})$ is surjective



$$\text{Map}(\mathbf{f}) \wedge \mathbf{R} \mathbf{f} \sqsubseteq \mathbf{F}(\mathbf{f}) \alpha \wedge \mathbf{1} \sqsubseteq \mathbf{R}^\circ \mathbf{R} \wedge \text{inductive}(\alpha^\circ \text{member}) \Rightarrow \mathbf{f}^\circ = (\mathbf{R})$$

(6.5g)

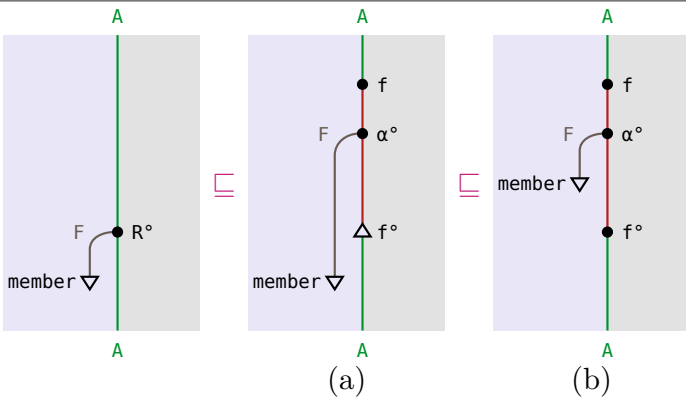
a map $\mathbf{f}$ with $\mathbf{R} \mathbf{f} \sqsubseteq \mathbf{F}(\mathbf{f}) \alpha$, for a surjective $\mathbf{R}$, has the fold $(\mathbf{R})$ as its converse



$$\text{Map}(\mathbf{f}) \wedge \mathbf{R} \mathbf{f} \sqsubseteq \mathbf{F}(\mathbf{f}) \alpha \Rightarrow \mathbf{R}^\circ \text{member} \sqsubseteq \mathbf{f} \alpha^\circ \text{member} \mathbf{f}^\circ$$

(6.5h)

$\mathbf{R}^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by $\mathbf{f}$, so it is inductive when $\alpha^\circ \text{member}$ is

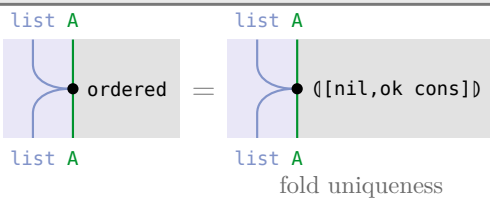


§6.6. Sorting by selection

$$\text{ordered} = ([\text{nil}, \text{ok cons}])$$

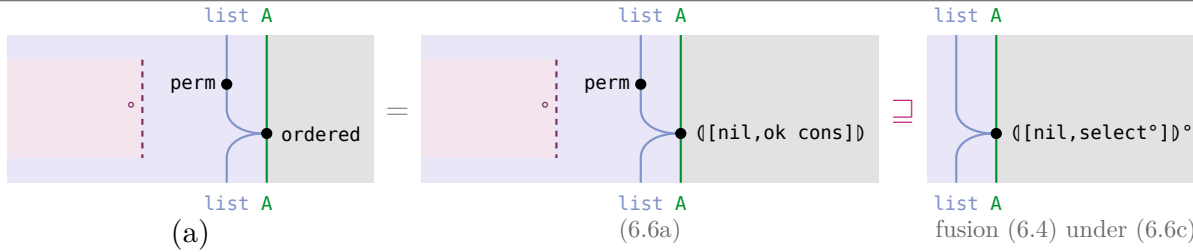
(6.6a)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it



$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq \text{perm ordered} \quad (6.6b)$$

every output of unfolding the input by **select** is a sorted permutation of it

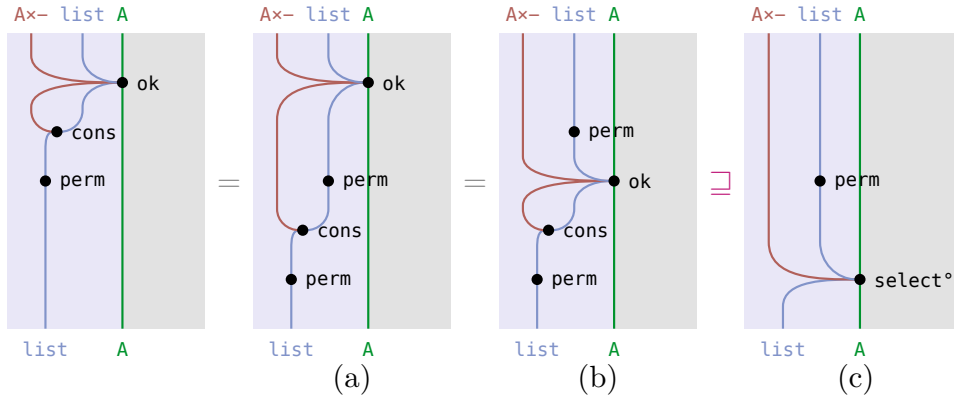


(a)

$\text{perm}^\circ = \text{perm}$, $\text{ordered}^\circ = \text{ordered}$

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



(a)

(b)

(c)

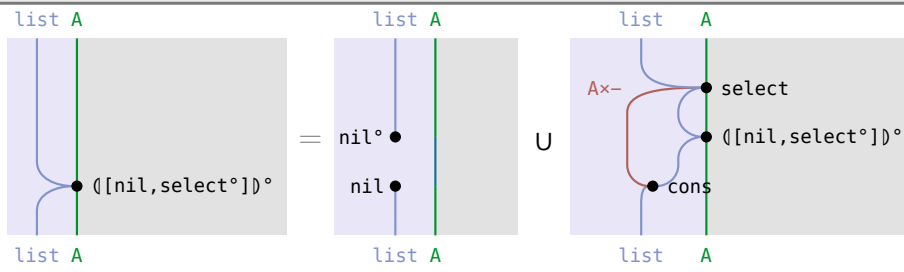
(a) $\text{cons perm} = (1 \times \text{perm}) \text{cons perm}$

(b) Ex 6.22, $\text{ok}(1 \times \text{perm}) = (1 \times \text{perm}) \text{ok}$

(c) $\text{select}^\circ \sqsubseteq \text{ok cons perm}$

$$([\text{nil}, \text{select}^\circ])^\circ = \text{nil}^\circ \text{nil} \cup \text{select} (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

selection sort returns [] on [], and otherwise selects (a, y), sorts y and conses a back on

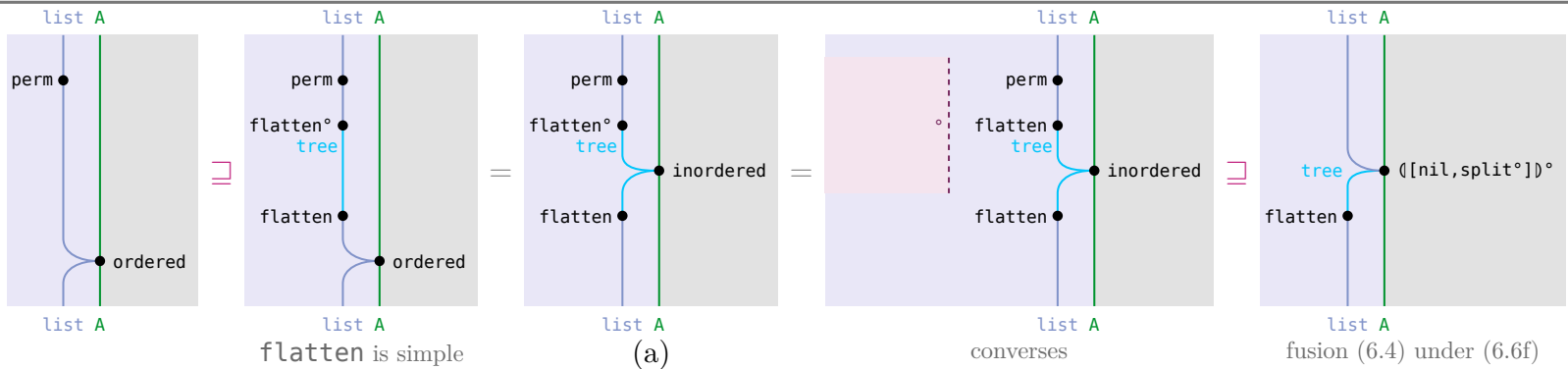


unfolding the converse of a fold

$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow \quad (6.6e)$$

$$([\text{nil}, \text{split}^\circ])^\circ \sqsubseteq \text{flatten} \sqsubseteq \text{perm ordered}$$

every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



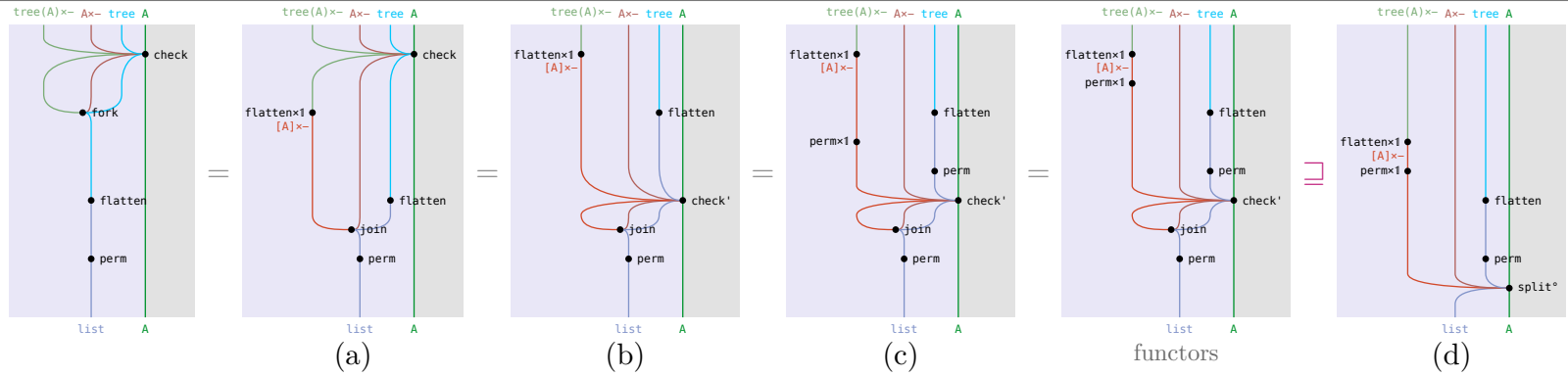
flatten is simple

(a)

(a) $\text{flatten ordered} = \text{inordered flatten}$

$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ \sqsubseteq \text{check fork flatten perm} \quad (6.6f)$$

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



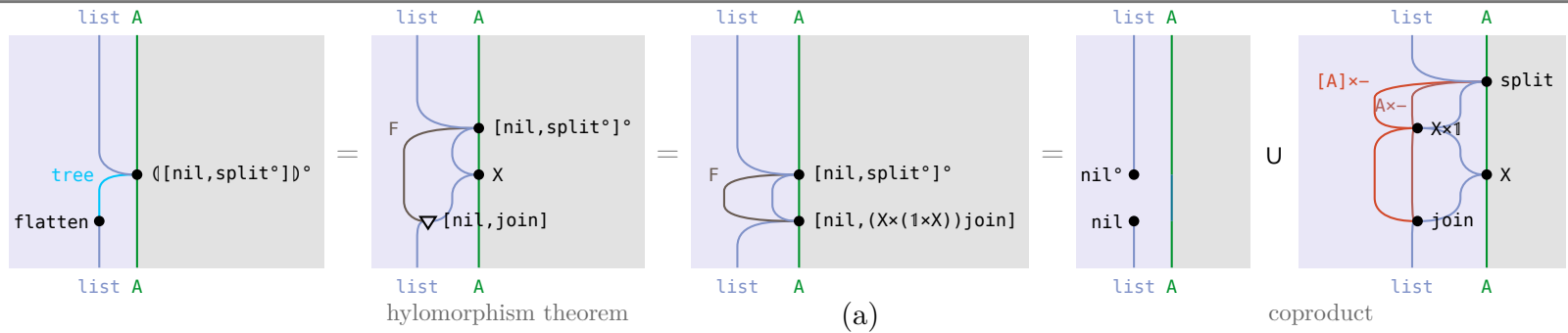
- (a) fork flatten = F(flatten)join
 (b) check F(flatten) = F(flatten)check'
 (c) join perm = F(perm)join perm, check' F(perm) = F(perm)check'
 (d) split^o ⊆ check' join perm

$$(\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{flatten} = \text{nil}^\circ \text{nil} \cup \text{split}$$

(6.6g)

$$(((\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{flatten}) \times (1 \times (\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{flatten}))) \text{join}$$

quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a



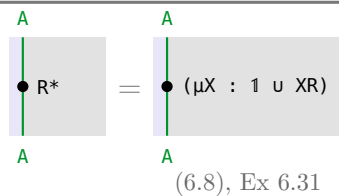
- (a) F(X)[nil, join] = [nil, (X × 1 × X) join]

§6.7. Closure

$$R^* = (\mu X : 1 \cup XR)$$

(6.7a)

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R***



(6.8), Ex 6.31

$$1 \sqsubseteq X \wedge XX \sqsubseteq X \Rightarrow REX \Leftrightarrow R^*EX$$

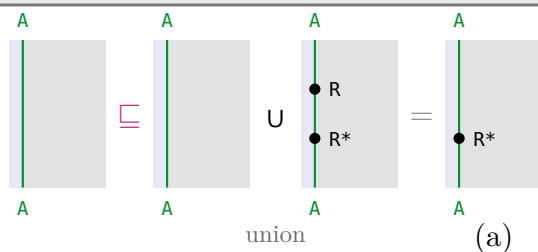
(6.7b)

for a preorder **X**, **X** contains **R** exactly when it contains **R***

$$1 \sqsubseteq R^*$$

(6.7c)

R* is reflexive



- (a) fixed point, Thm 6.1

(6.7d)

$R \subseteq R^*$
 R^* contains R

identity
(a) $1 \subseteq R^*$ — (6.7c)
(b) fixed point

(6.7e)

$R^* R^* \subseteq R^*$
 R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup RX$

(a) composition distributes over $\cup$
(b) $RR^* \subseteq R^*$ — (6.7d)

(6.7f)

$1 \in X \wedge XX \subseteq X \wedge R \in X \Rightarrow R^* \in X$
a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup RX$, so it contains R^*

$1 \in X, R \in X$ $XX \subseteq X$

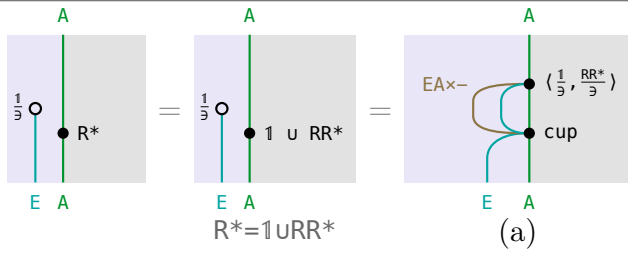
(6.7g)

$\text{inductive}(R) \wedge X = 1 \cup RX \Rightarrow X = R^*$
when R admits induction, R^* is the only X equal to $1 \cup RX$

(a)	$((R^* \Rightarrow R^*/X)/R) \cap R^*)X$	
(b)	$((R^* \Rightarrow R^*/X)/R) \cap R^*) (1 \cup RX)$	$X = 1 \cup RX$
(c)	$((R^* \Rightarrow R^*/X)/R) \cap R^* \cup ((R^* \Rightarrow R^*/X)/R) \cap R^*)RX$	$R(SuT) = RSuRT$ — composition distributes over $\cup$
(d)	$R^* \cup ((R^* \Rightarrow R^*/X) \cap R^*)X$	$a \subseteq c, b \subseteq d \Rightarrow a \cup b \subseteq c \cup d$ — $\cup$ is monotone; $(Z/R)R \subseteq Z$, $R^*R \subseteq R^*$ justify the right side
(e)	R^*	$(R^* \Rightarrow R^*/X) \cap R^* \subseteq R^*/X$

$$\frac{R^*}{S} = \left(\frac{1}{S}, \frac{RR^*}{S} \right) \text{cup} \quad (6.7h)$$

the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors



$$(a) \ \wedge(R \cup S) = \langle \wedge(R), \wedge(S) \rangle \text{cup}$$

$$R - \emptyset = R \quad (6.7i)$$

taking nothing away leaves R

$$R \cup (S - R) = R \cup S$$

beside R , only the part of S outside R adds anything

$$R - (S \cup T) = R - S - T$$

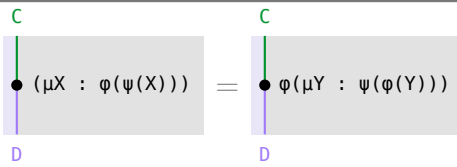
taking away S and then T takes away $S \cup T$

$$(R \cup S) - T = (R - T) \cup (S - T)$$

taking T away from a union takes it away from each part

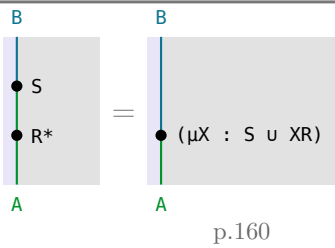
$$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y))) \quad (6.7j)$$

the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ



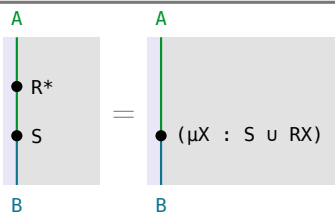
$$SR^* = (\mu X : S \cup XR) \quad (6.7k)$$

S followed by any number of R steps is the least X containing S and closed under a further R step



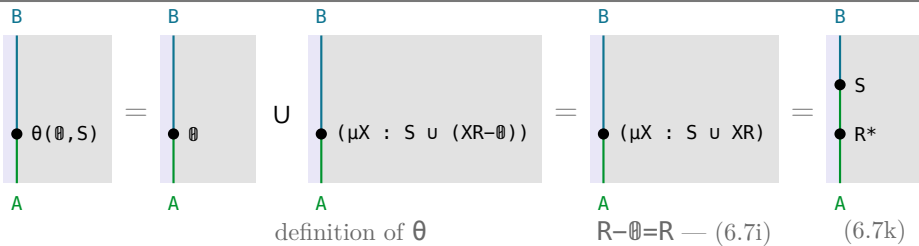
$$R^*S = (\mu X : S \cup RX)$$

any number of R steps followed by S is the least X containing S and closed under an R step in front



$$\theta(\emptyset, S) = SR^* \quad (6.7l)$$

started with nothing found and S to explore, θ computes SR^*



$\theta(P, \emptyset) = P$ (6.7m)

with nothing left to explore, θ returns what it has found

definition of θ

(a) $\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$

$\theta(P, Q) = \theta(P \cup Q, QR - P - Q)$ (6.7n)

one round moves Q into the found part and explores the new R -successors of Q that are in neither

(a) (b) (c)

(a)	$\theta(P, Q)$	
(b)	$P \cup (\mu X : Q \cup (XR - P))$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup (XR - P))$ — definition of θ
(c)	$P \cup (\mu X : Q \cup (XR - P - Q))$	$Q \cup S = Q \cup (S - Q)$ — (6.7i)

(a)

(a)	$P \cup (Q \cup (\mu X : (Q \cup X)R - P - Q))$	$(\mu X : \phi(\psi(X))) = \phi(\mu Y : \psi(\phi(Y)))$ — (6.7j)
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(a)

(b)

(a)	$P \cup Q \cup (\mu X : (QR - P - Q) \cup (XR - (P \cup Q)))$	$((Q \cup X)R - P) - Q = (QR - P) - Q \cup (XR - (P \cup Q))$ — composition distributes over $\cup$, then subtraction distributes, (6.7i)
(b)	$\theta(P \cup Q, QR - P - Q)$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup (XR - P))$ — definition of θ